Kaelan Patterson

Mississauga, ON | 647-202-2637 | kpatte16@uoguelph.ca
Kaelan's Portfolio | LinkedIn

Skills Summary

- Hands-on mechanical assembly experience from separate VEX robot and Arduino-based teddy bear wheelchair projects.
- C programming and JavaScript/Node.js development through game and automation projects; introductory Python coursework.
- Practical experience with REST APIs, webhooks and data storage using Express, Twilio, OpenAI, Supabase and Google Places.
- Basic AutoCAD skills developed through room layout plans and orthographic drawings.
- Troubleshooting experience identifying drivetrain failures in a robot and investigating API calls, filtering logic and data persistence in software prototypes.
- Communication, teamwork and initiative developed through founding Voxa Automations, presenting demonstrations, robotics projects and hockey officiating.

Education

Bachelor of Engineering, Mechatronics Engineering (Co-op) 2025 - Present
University of Guelph, Guelph, ON

- Second-year student; expected graduation in 2030.

Ontario Secondary School Diploma 2021 - 2025
St. Marcellinus C.S.S., Mississauga, ON

- Honour Roll student in Grades 9-12.
- Robotics Team Vice President: taught Arduino basics and contributed mechanical work and code to a VEX robot that completed its assigned task.

Academic Projects

Student Designer and Builder, Arduino Wheelchair 2025
University of Guelph, Guelph, ON

- Built the drivetrain and launcher arm for an Arduino-based teddy bear wheelchair, integrating the mechanical subsystems into a physical prototype.
- Planned component placement to fit the subsystems on the robot while accounting for the centre of mass and the space available.
- Connected motors and wired components using a breadboard and Arduino board during assembly of the teddy bear wheelchair.
- Replaced thin elastic drivetrain bands that broke during testing with a thicker band and adjusted its position to increase tension; the robot completed both required tasks on test day.

Student Programmer, Tic-Tac-Toe Game 2025
University of Guelph, Guelph, ON

- Developed a tic-tac-toe game in C for a programming project, implementing friend and computer modes to support multiplayer and single-player use.
- Programmed computer move selection with selectable difficulty, giving single-player users a choice of challenge levels.
- Implemented configurable 3 x 3 to 10 x 10 boards, allowing players to change the board size within one program.

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Technical Experience

Founder and Automation Developer April - August 2026
Voxa Automations, Mississauga, ON

- Built an AI receptionist prototype for business inquiries using Node.js, Twilio and OpenAI APIs, enabling callers to describe their needs and leave requests for staff follow-up.
- Connected call-handling webhooks to Supabase storage and SMS notifications, turning phone inquiries into saved leads and staff alerts within one workflow.
- Developed Voxa Lead Finder with JavaScript, Node.js/Express and Google Places API, enabling users to find businesses by type and location and filter by website availability.
- Implemented lead scoring, call-status tracking, notes and Excel export; investigated filtering and data-persistence issues while refining the prototype.
- Created a property-management call simulation and booking pages, giving prospective users a demonstration of request intake, urgency checks and on-call escalation.

Work and Volunteer Experience

Hockey Referee November 2025 - Present
Ontario Minor Hockey Association, Guelph, ON

- Assess fast-moving gameplay and apply league rules, making time-sensitive decisions that keep games progressing under consistent standards.
- Coordinate with fellow officials on complex calls and explain rulings to players and coaches, establishing a shared decision before play resumes.

Hockey Referee July 2024 - Present
Canlan Classic Tournaments, Etobicoke, ON

- Monitor player conduct and protective equipment during tournaments, addressing unsafe behaviour to support compliance with league safety requirements.

Hockey Timekeeper September 2024 - October 2025
Streetsville Derbys Junior Hockey Club, Mississauga, ON

- Set up and tested scorekeeping, announcing and audio equipment before games, preparing the timing and communication systems for play.
- Tracked game time, goals, penalties and player statistics, then compiled scoresheets and post-game reports to produce a complete official game record.

Hockey Referee and Timekeeper October 2021 - September 2025
Streetsville Hockey League, Mississauga, ON

- Applied rules, tracked game events and communicated decisions while maintaining composure under pressure, helping games run safely and consistently.

Head Football Coach, U12 (Volunteer) June - August 2023
UA Flag Football Canada, Mississauga, ON

- Coached and mentored U12 players in football fundamentals, game awareness and teamwork, building foundational skills for future competition while leading the team to the finals.

Certifications

WHMIS Certification

Hockey Canada Level 2 Official